

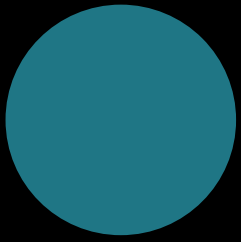


BEST PRACTICES FOR:

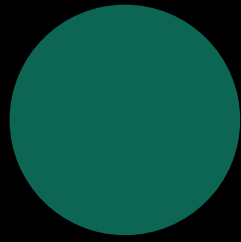
Gene family reconstruction

Dr. Laurence Belcher

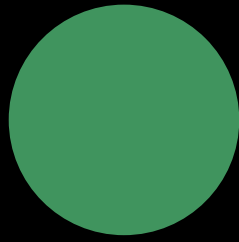
FIVE QUESTIONS



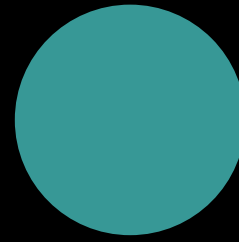
HOW DO I DECIDE
WHICH SPECIES TO
INCLUDE?



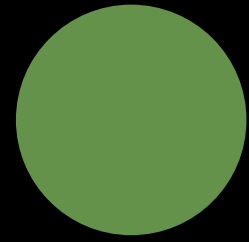
WHAT AM I TRYING
TO INFER?



HOW MUCH
SHOULD I TRUST
MY
ORTHOGROUPS?

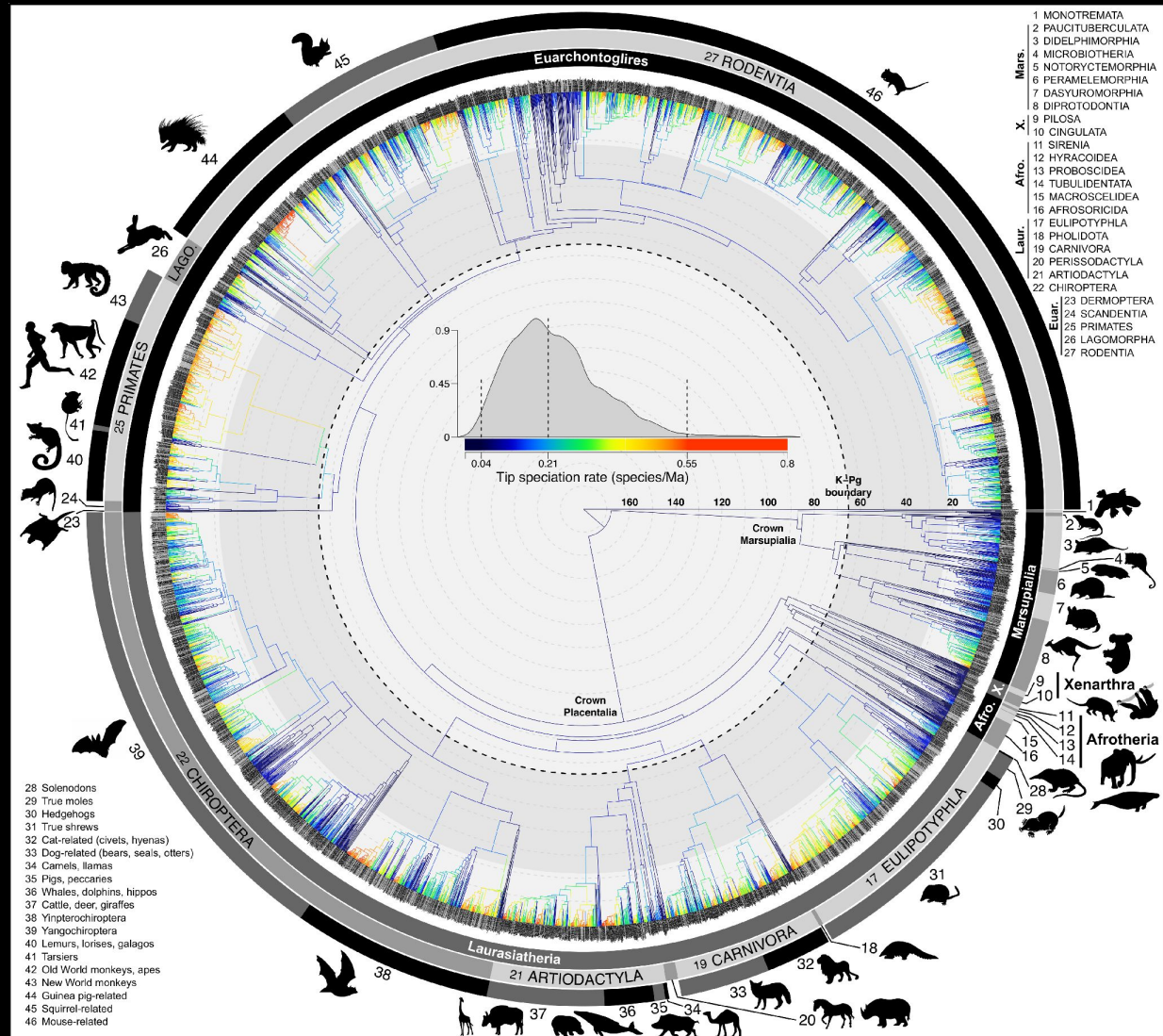


SHOULD I REFINE
MY ORTHOGROUPS
AT THE END?



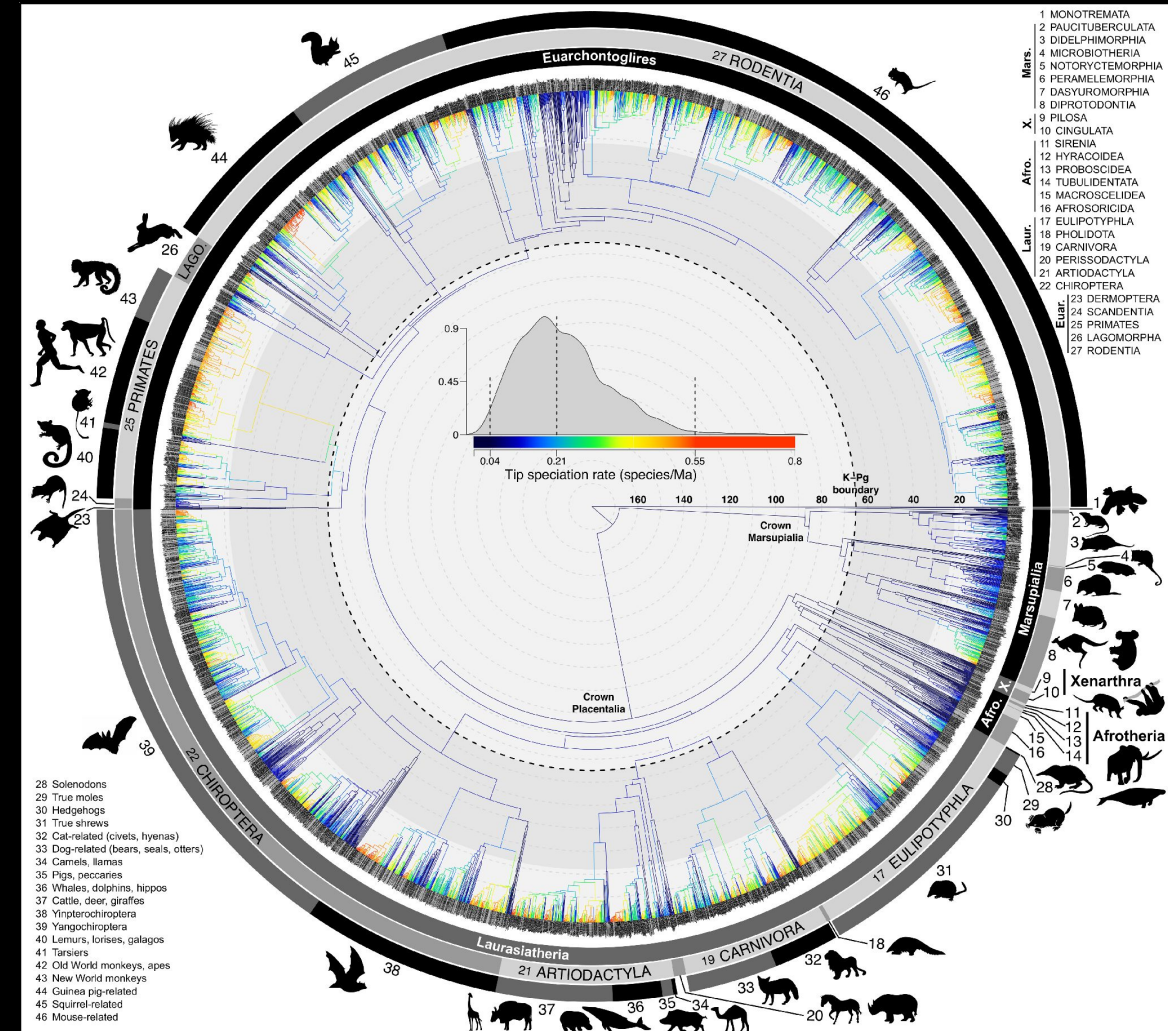
RECORD WHAT
YOU DID

HOW DO I DECIDE WHICH SPECIES TO INCLUDE?



HOW DO I DECIDE WHICH SPECIES TO INCLUDE?

- Optimise phylogenetic diversity, taxonomic coverage, and priority species
- Same logic for scalable methods like OrthoFinder –core –assign to choose the core species
- Don't always need an outgroup, but if you do then pick one close yet unambiguously outside tree
- Many methods will make a species tree for you, but if you have a good one then use it

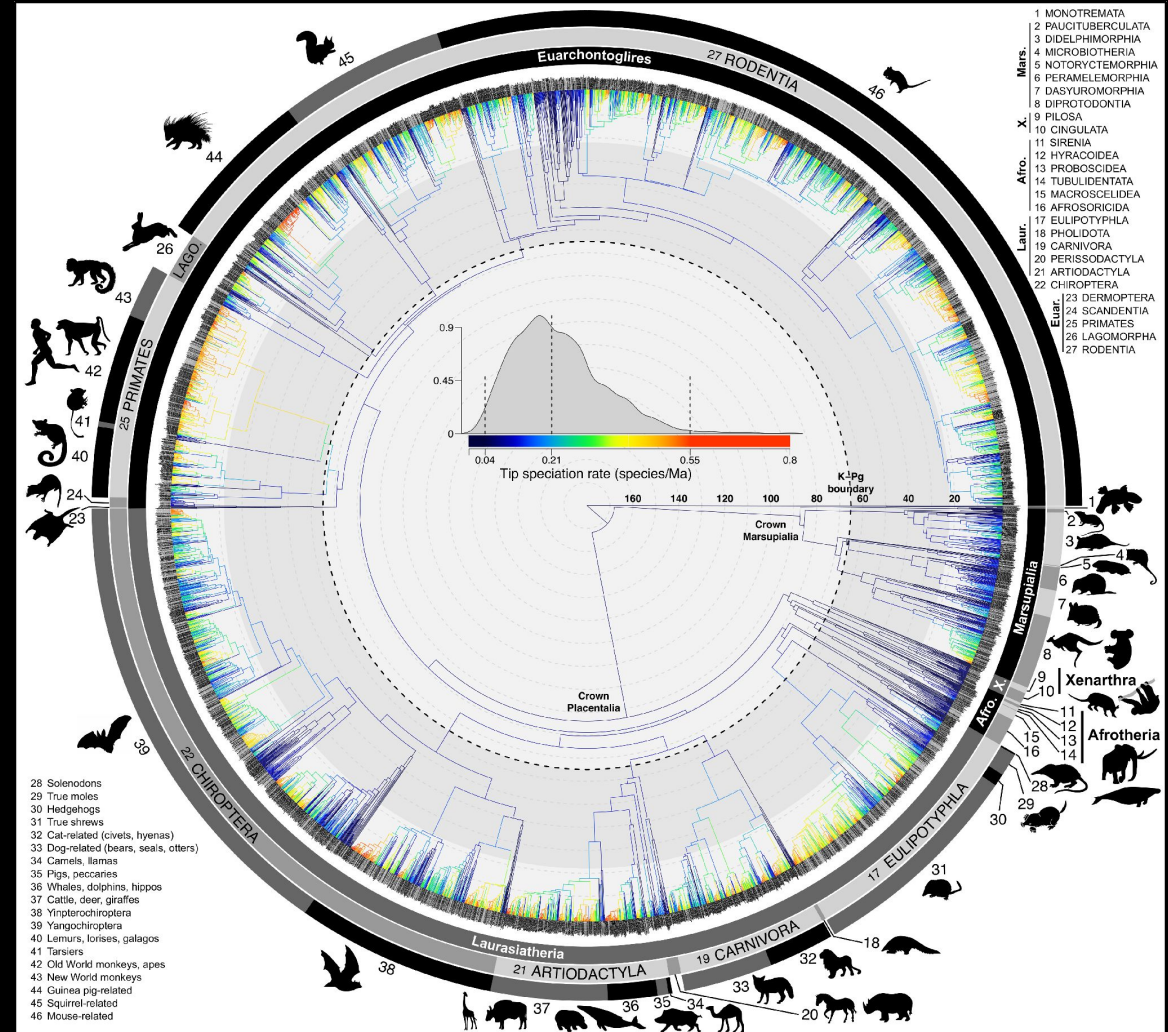


HOW DO I DECIDE WHICH SPECIES TO INCLUDE?

- 1) Prune the phylogeny to species with proteomes (~400 species)
- 2) Decide how many species you want to analyse (e.g. 100)
- 3) Identify key species that you definitely want (e.g. Giant anteater *M. tridactyla*)
- 4) Run PDA (Phylogenetic Diversity Analyser) to pick remaining n=99 species
- 5) Prune the tree to those 100 species

Depending on tool:

- 6) Add an outgroup (e.g. chicken)
- 7) Run PDA to get a core set of n=32 species



WHAT AM I TRYING TO INFER?

- Orthogroup
- Pairwise orthologs
- Single-copy orthogroups
- HOGs at a taxonomic level

WHAT AM I TRYING TO INFER?

Orthogroup

-> to look at expansions/contractions

Pairwise orthologs

-> for direct functional transfer

Single-copy orthogroups

-> for making species tree or dN/dS scans

HOGs at a taxonomic level

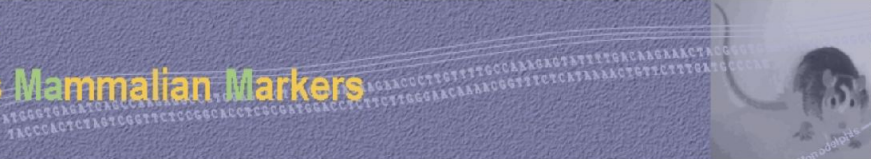
-> fixed answer regardless of adding or removing species

WHAT AM I TRYING TO INFER?

- Check what you are getting from the tool you choose
- Check if you can avoid computation e.g. from existing databases



A database of
Orthologous Mammalian Markers



Description :

The **NCBI** database was used to define a set of **1-to-1 orthologous genes** from available mammalian genomes from which annotated CoDing Sequences (CDSs) were extracted. The phylogenetic signal and evolutionary characteristics of these single-copy CDSs were then evaluated using a bioinformatics pipeline implementing state-of-the-art tools for sequence alignment (**MAFFT** + **MACSE**), alignment cleaning (**HMMCleaner** + **PhylteR**), and phylogenetic inference (**IQ-TREE** + **ERaBLE**). The resulting OrthoMam database of curated nucleotide and amino acid alignments and corresponding phylogenetic trees can be interrogated through this website.

NUMBER OF SPECIES

190

↑ 63% Since last version





HOW MUCH SHOULD I TRUST MY ORTHOGROUPS?

Consider validating against a gene family you already know

-> e.g. something well characterized in your species

Know that errors are not random

-> e.g. orthology inference is harder for deep divergence, rapid evolution, fissions and fusions

Most tools are designed to be easy to use, but allow customization for advanced usage

-> e.g. changing MCL inflation factor makes orthogroups smaller or larger

SHOULD I REFINE MY ORTHOGROUPS AT THE END?

Options include:

Synteny

- Using the idea that conserved gene order distinguishes true orthologs from paralogs in multi-copy groups
- Implemented in orthoSynAssign (Tsai *et al.* (2026))

Domain architecture

- Using the idea that sequence similarity and domain architecture can diverge independently, so that high similarity doesn't guarantee conserved domains
- Implemented with FAS score (annoFAS)

RECORD WHAT YOU DID

Version number of tool

Exact command you ran

Exact input data you
provided

CONCLUSION

01

Choose your species
carefully

02

Know which object
you want

03

Inspect where
evidence is weakest

04

Bring in post-hoc
things like synteny
when its helpful

05

Record what you did